

Fall 2025 DRP Presentation: Quadratic Reciprocity

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Motivation

When working in the ring $\mathbb{Z}/p\mathbb{Z}$ one natural question one might have would be 'for some $a \in \mathbb{Z}/p\mathbb{Z}$, is a a square'? Or, more formally,

For some $a \in \mathbb{Z}/p\mathbb{Z}$ does there exist a solution to the congruence

$$x^2 \equiv a \pmod{p}$$

For example,

- Is 2 a square mod 7?
- Is 11 a square mod 71?
- Is 41 a square mod 67?

Background (Algebraic Integers)

Definition

An **algebraic integer** is a complex number ω that is the root of a polynomial $x^n + a_1x^{n-1} + \cdots + a_n = 0$, where $a_1, a_2, \dots, a_n \in \mathbb{Z}$.

Theorem

The set of algebraic integers forms a ring.

When working in the ring of algebraic integers, Ω , we define congruence as follows. If $\omega_1, \omega_2, \gamma, \alpha \in \Omega$, we say that $\omega_1 \equiv \omega_2 \pmod{\gamma}$ if $\omega_1 - \omega_2 = \gamma\alpha$. Congruence in Ω follows all the standard properties that hold in $\mathbb{Z}$. We also have the following that is true in Ω : If $\omega_1, \omega_2 \in \Omega$ and $p \in \mathbb{Z}$ is a prime, then

$$(\omega_1 + \omega_2)^p \equiv \omega_1^p + \omega_2^p \pmod{p}.$$

This fact can be easily proven by expansion with the binomial theorem and the fact that $p \mid \binom{p}{k}$ for $1 \leq k \leq p-1$.

Background (Legendre Symbol)

Definition

Let p be a prime and a any integer. We then define

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square mod } p \\ -1 & \text{if } a \text{ is not a square mod } p \\ 0 & \text{if } p \mid a \end{cases}$$

as the **Legendre Symbol**.

Properties of the Legendre Symbol

- 1 $a^{p-1/2} \equiv (a/p) \pmod{p}$
- 2 $(ab/p) = (a/p)(b/p)$
- 3 If $a \equiv b \pmod{p}$ then $(a/p) = (b/p)$

The Law of Quadratic Reciprocity

With the following background established we can now state the Law of Quadratic Reciprocity:

Theorem (Quadratic Reciprocity)

For two odd primes p and q with $p \neq q$

$$\left(\frac{p}{q}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} \left(\frac{q}{p}\right)$$

Notice the following:

- p **or** $q \equiv 1 \pmod{4} \Rightarrow (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} = 1$
- p **and** $q \equiv 3 \pmod{4} \Rightarrow (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} = -1$

We also have the following known Legendre Symbol expressions:

$$(-1/p) = (-1)^{(p-1)/2}$$

$$(2/p) = (-1)^{(p^2-1)/8}$$

Foundations of the Proof (Roots of Unity)

Definition

An n -th **root of unity** is any value that is a solution to the equation

$$x^n - 1 = 0$$

For the rest of this discussion we shall let $\zeta = e^{2\pi i/p}$ which is a p -th root of unity. The following lemma is an important fact about ζ that we will use in our proof.

Lemma (Orthogonality of Characters)

The sum $\sum_{t=0}^{p-1} \zeta^{at}$ equals p if $a \equiv 0 \pmod{p}$ and is 0 otherwise.

Proof.

If $a \equiv 0 \pmod{p}$ then $\zeta^a = 1$ by definition of a root of unity. It follows that $\sum_{t=0}^{p-1} \zeta^{at} = p$. If $a \not\equiv 0 \pmod{p}$ then $\zeta^a \neq 1$ and $\sum_{t=0}^{p-1} \zeta^{at} = (\zeta^{ap} - 1)(\zeta^a - 1) = 0$. □

Foundations of the Proof (Quadratic Gauss Sums)

Definition

A quadratic gauss sum, g_a is defined as

$$g_a = \sum_{t=0}^{p-1} \left(\frac{t}{p} \right) \zeta^{at}$$

with g_1 commonly notated as g .

It is important to note that $g_a = (a/p)g$.

Lemma

$$g^2 = (-1)^{(p-1)/2} p$$

Proof of Lemma

To prove the lemma we will consider the sum $\sum_{a=0}^{p-1} g_a g_{-a}$ in two different ways and our result will follow from their equality. First, we will leverage the identity $g_a = (a/p) g$ to show that

$$\sum_{a=0}^{p-1} g_a g_{-a} = \left(\frac{-1}{p} \right) (p-1) g^2.$$

Next, we will rewrite the product $g_a g_{-a}$ using the definition of a Gauss Sum, sum over a , and apply the Orthogonality of Characters:

$$\sum_a g_a g_{-a} = \sum_a \sum_x \sum_y \left(\frac{x}{p} \right) \left(\frac{y}{p} \right) \zeta^{a(x-y)} = (p-1)p$$

So,

$$\left(\frac{-1}{p} \right) (p-1) g^2 = (p-1)p \Rightarrow g^2 = \left(\frac{-1}{p} \right) p$$



Proof of the Law of Quadratic Reciprocity

With this information now in mind we can start with the proof. Consider $p^* = (-1)^{(p-1)/2} p = g^2$, and another odd prime q such that $q \neq p$. We will be working with congruences mod q in the ring of algebraic integers:

$$g^{q-1} = (g^2)^{(q-1)/2} = p^{*(q-1)/2} \equiv \left(\frac{p^*}{q}\right) \pmod{q}$$

$$\Rightarrow g^q \equiv \left(\frac{p^*}{q}\right) g \pmod{q}$$

Rewriting g^q using the definition of the Gauss sum gives

$$g^q = \left(\sum \left(\frac{t}{p}\right) \zeta^t\right)^q \equiv \sum \left(\frac{t}{p}\right)^q \zeta^{qt} \equiv g_q \pmod{q}$$

Recalling that $g_a = (a/p)g$, we now have that $g^q \equiv g_q \equiv (q/p)g \pmod{q}$, thus

$$\left(\frac{q}{p}\right) g \equiv \left(\frac{p^*}{q}\right) g \pmod{q}$$

Proof (cont.)

If we multiply both sides by g and apply the fact that $g^2 = p^*$ we have

$$\begin{aligned}\left(\frac{q}{p}\right) p^* &\equiv \left(\frac{p^*}{q}\right) p^* \pmod{q} \\ \Rightarrow \left(\frac{q}{p}\right) &\equiv \left(\frac{p^*}{q}\right) \pmod{q} \\ \Rightarrow \left(\frac{q}{p}\right) &= \left(\frac{p^*}{q}\right)\end{aligned}$$

Which was to be shown. □

Worked Example

Earlier I asked if 11 was a square, or quadratic residue, mod 71. With the law of quadratic reciprocity we can now figure this out:

$$\begin{aligned}\left(\frac{11}{71}\right) &= (-1)^{(11-1/2)(71-1/2)} \left(\frac{71}{11}\right) \\ &= - \left(\frac{5}{11}\right) = - (-1)^{(5-1/2)(11-1/2)} \left(\frac{11}{5}\right) \\ &= - \left(\frac{1}{5}\right) = - (-1)^{5-1/2} = -1\end{aligned}$$

Therefore, 11 **is not** a square mod 71. This also implies that 71 **is** a square mod 11.

References



K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd Edition, Springer, New York, 1990.